Chapter 1

1.5 Circle

Completing the Square

a) $x^2 + bx + c = 0$

$$\underbrace{x^2 + bx + \left(\frac{b}{2}\right)^2} - \left(\frac{b}{2}\right)^2 + c = 0$$

$$\left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c = 0$$

b) $x^2 - bx + c = 0$

$$\underbrace{x^2 - bx + \left(\frac{-b}{2}\right)^2} - \left(\frac{-b}{2}\right)^2 + c = 0$$

$$\left(x - \frac{b}{2}\right)^2 - \left(\frac{-b}{2}\right)^2 + c = 0$$

Note: The coefficients of x^2 must be 1.

Example 1

Perform completing the square for each following equation.

a) $x^2 - 4x - 3 = 0$

b) $2x^2 + 10x + 7 = 0$

c) $3x^2 + 3y^2 - 18x + 6y - 45 = 0$

Solution

a) $x^2 - 4x - 3 = 0$

$x^2 - 4x + (\ \ )^2 - (\ \ )^2 - 3 = 0$

$$\underbrace{x^2 - 4x + \left(\frac{-4}{2}\right)^2} - \left(\frac{-4}{2}\right)^2 - 3 = 0$$

$$(x-2)^2 - (-2)^2 - 3 = 0$$

$$(x-2)^2 - 7 = 0$$

$$\therefore \quad (x-2)^2 = 7$$

b) $2x^2 + 10x + 7 = 0$

$\div 2$: $x^2 + 5x + \frac{7}{2} = 0$

The coefficient of x^2 must be 1

$x^2 + 5x + (\ )^2 - (\ )^2 + \frac{7}{2} = 0$

$\underline{x^2 + 5x + \left(\frac{5}{2}\right)^2} - \left(\frac{5}{2}\right)^2 + \frac{7}{2} = 0$

$\left(x + \frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2 + \frac{7}{2} = 0$

$\left(x + \frac{5}{2}\right)^2 - \frac{11}{4} = 0$

$\left(x + \frac{5}{2}\right)^2 = \frac{11}{4}$

c) $3x^2 + 3y^2 - 18x + 6y - 45 = 0$

$\div 3$: $x^2 + y^2 - 6x + 2y - 15 = 0$

$x^2 - 6x + (\ )^2 - (\ )^2 + y^2 + 2y + (\ )^2 - (\ )^2 - 15 = 0$

$x^2 - 6x + \left(\frac{-6}{2}\right)^2 - \left(\frac{-6}{2}\right)^2 + y^2 + 2y + \left(\frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 - 15 = 0$

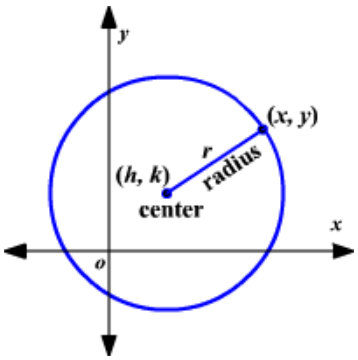
$\underline{x^2 - 6x + (-3)^2} - (-3)^2 + \underline{y^2 + 2y + (1)^2} - (1)^2 - 15 = 0$

$(x-3)^2 - (9) \quad + (y+1)^2 - 1 \quad - 15 = 0$

$(x-3)^2 + (y+1)^2 - 25 = 0$

$(x-3)^2 + (y+1)^2 = 25$

Circles



The standard form of the equation of a circle is

$$(\boldsymbol{x-h})^2 + (\boldsymbol{y-k})^2 = \boldsymbol{r^2}$$

Where (h, k) = centre and r = radius

Example 2

Find the centre and radius of the equation of a circle as below.

$$(x+5)^2+(y-1)^2=4$$

Solution

$x+5=0 \quad$ and $\quad y-1=0$
$x=-5 \quad$ and $\quad y=1$
$\therefore$ centre $=(-5,1)$

$r^2=4$
$\therefore$ radius, $r=2$

Example 3

Write the equation of a circle $x^2-14x+y^2+8y+40=0$ in standard form. Hence, sketch the circle by indicating its centre and radius.

Solution

$$x^2-14x+(\ )^2-(\ )^2+y^2+8y+(\ )^2-(\ )^2+40=0$$
$$x^2-14x+\left(\frac{-14}{2}\right)^2-\left(\frac{-14}{2}\right)^2+y^2+8y+\left(\frac{8}{2}\right)^2-\left(\frac{8}{2}\right)^2+40=0$$

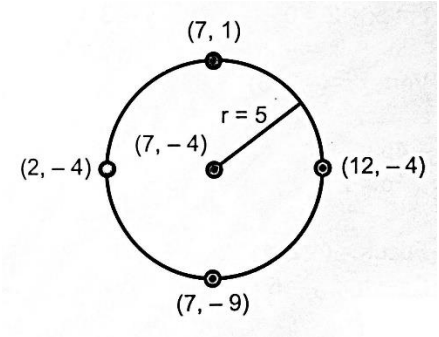
$$\underbrace{x^2-14x+(-7)^2}-(-7)^2+\underbrace{y^2+8y+(4)^2}-(4)^2+40=0$$

$$(x-7)^2-(49)+(y+4)^2-(16)+40=0$$
$$(x-7)^2+(y+4)^2-25=0$$
$$(x-7)^2+(y+4)^2=25$$

Therefore

$x-7=0 \quad$ and $\quad y+4=0$
$x=7 \quad$ and $\quad y=-4$
$\therefore$ centre $=(7,-4)$

$r^2=25$
$\therefore$ radius, $r=5$



Example 4

Find the equation of a circle with centre $(-3, 1)$ and passes through the point $(-1, 4)$.

Solution

$$\text{Radius, } r = \text{ Distance between 2 points}$$
$$= \sqrt{[-1-(-3)]^2 + (4-1)^2}$$
$$= \sqrt{(2)^2 + (3)^2}$$
$$= \sqrt{13}$$

$\therefore r = \sqrt{13}$ and centre $= (-3, 1)$

From the standard form of circle: $(\boldsymbol{x-h})^2 + (\boldsymbol{y-k})^2 = \boldsymbol{r^2}$

$$[x-(-3)]^2 + (y-1)^2 = \left(\sqrt{13}\right)^2$$
$$(x+3)^2 + (y-1)^2 = 13$$
$$(x+3)^2 + (y-1)^2 = 13$$
$$x^2 + 6x + 9 + y^2 - 2y + 1 - 13 = 0$$
$$x^2 + y^2 + 6x - 2y - 3 = 0$$

General form: $Ax^2 + By^2 + Cy + Dx + C = 0$

Example 5

Find the equation of a circle with the points $A(3, 5)$ and $B(7, 1)$ as the ends of its diameter.

Solution

$$\text{Centre} = \text{Midpoint of AB}$$

$$= \left(\frac{3+7}{2}, \frac{5+1}{2}\right)$$
$$= \left(\frac{10}{2}, \frac{6}{2}\right)$$
$$= (5, 3)$$

$$\text{Radius, r} = \text{Distance of AB}$$

$$= \sqrt{(7-5)^2 + (1-3)^2}$$
$$= \sqrt{(2)^2 + (-2)^2}$$
$$= \sqrt{8}$$

From the standard form of circle $(\boldsymbol{x-h})^2 + (\boldsymbol{y-k})^2 = \boldsymbol{r^2}$

$$(x-5)^2 + (y-3)^2 = \left(\sqrt{8}\right)^2$$
$$x^2 - 10x + 25 + y^2 - 6y + 9 = 8$$
$$x^2 + y^2 - 10x - 6y + 26 = 0$$